



Mark Scheme (Results)

October 2021

Pearson International Advanced
Subsidiary Level
In Chemistry (WCH13)
Paper 01: Practical Skills in Chemistry I

Question Number	Answer	Additional Guidance	Mark
1(a)	An answer that makes reference to the following points: - test (1) - colour (1)	Flame test Allow description of a flame test Ignore burning Red / crimson Allow scarlet Do not award brick red / yellow red / orange Allow sulfate (solution) added and forms cloudy solution/white ppt for 1 mark Marks are independent	(2)

Question Number	Answer	Additional Guidance	Mark
1(b)	An answer that makes reference to the following points: - correct result for barium chloride (1) - correct result for silver nitrate (1)	No visible change / no reaction Allow no results / no observation / no change / no precipitate formed Do not award "nothing" alone White and precipitate/ppt/solid/crystals Ignore darkens in sunlight Ignore dissolves in dilute ammonia Ignore insoluble in acid Do not award "soluble in excess" Do not award gas formed / effervescence	(2)

Question Number	Answer	Additional Guidance	Mark
1(c)	suitable suggestion relating to the high temperature required	e.g. Bunsen burners will not be hot enough, better equipment will be required to reach the correct temperature, etc. Allow temperature cannot be reached (by school equipment) / temperature is too high Allow a very high temperature is needed Ignore references to energy/heat/power Ignore mention of toxic gas Ignore mention of expense Ignore references to safety Ignore "lack of supplies" alone Do not award "will catch fire"	(1)

Question Number	Answer	Additional Guidance	Mark
1(d)(i)	An answer that makes reference to the following points: heat (and reweigh) to constant mass	Allow "no more brown gas given off" Allow "no more NO₂ given off" Allow "no longer relights a glowing splint" Ignore "no more gas/O₂ given off" Ignore "heat very strongly" Ignore references to high temperatures Ignore references to mass/volume of gas Do not award references to burning	(1)

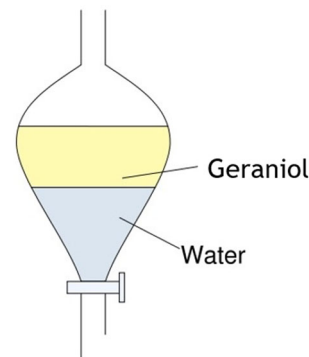
Question Number	Answer	Additional Guidance	Mark
1(d)(ii)	colour of NO₂	Brown	(1)

Question Number	Answer	Additional Guidance	Mark
1(d)(iii)	correct procedure and result	(Re)lights a glowing splint Allow "rekindles" or "ignites" Allow "smouldering" Allow splinter / stick / spill /description of a splint Do not award "pops" NB There must be some reference to the splint having been recently extinguished and containing embers.	(1)

Question Number	Answer	Additional Guidance	Mark
1(d)(iv)	An answer that makes reference to the following points: - observation - product	(1) Solid dissolves / forms a colourless solution Allow solid disappears Allow gets warmer Allow steam given off Allow sizzling sound Ignore bubbles/effervescence/fizzing (1) Strontium hydroxide / Sr(OH)₂ If name and formula given both must be correct Do not award hydrogen / oxygen Marks are independent	(2)

(Total for Question 1 = 10 marks)

Question Number	Answer	Additional Guidance	Mark
2(a)	An answer that makes reference to the following points: - alkene with correct test - correct colour change for alkene - alcohol with correct test - correct result	(1) Alkene and (shake with) bromine (water) Ignore C=C (1) Decolourises OR (brown / orange / yellow) to colourless Allow acidified (potassium) manganate (VII), (pink/purple) to colourless (1) Alcohol and add PCl₅ / phosphorus(V) chloride (or other accepted test for M3 and observation for M4, see below) Allow phosphorous pentachloride Allow hydroxy(l) Ignore -OH Do not award hydroxide Do not award PCl₃ (1) Misty fumes Allow white fumes / steamy fumes Allow fumes turn damp blue litmus red Accepted tests <u>with</u> named alcohol group: Heat with acidified potassium dichromate((VI)) (Orange) to green / blue Add sodium Bubbles / effervescence Add a carboxylic acid and a strong acid A fruity smell M2 and M4 are dependent on the correct test for each being given in M1 and M3 even if the mark is not awarded	(4) Expert

Question Number	Answer	Additional Guidance	Mark
2(b)	An answer that makes reference to the following points: • separating funnel (1) • diagram of separating funnel (1)	Example of a diagram: Allow separatory/separation funnel  Diagram should show a tap - but it does not need to be labelled - and a narrow top of the vessel, capable of being stoppered Ignore stoppers/closed top Do not award M2 if tap is labelled as a stopper Geraniol labelled as the top layer Allow organic layer / alcohol layer	(3) Expert
	• layers and the right way around (1)		

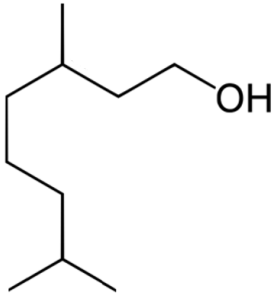
Question Number	Answer	Additional Guidance	Mark
2(c)	An answer that makes reference to the following points: - drying agent - description of drying: mixing - description of drying: separating	Ignore distillation (1) Named substance / formula (Anhydrous) calcium chloride / CaCl_2 (Anhydrous) sodium sulfate / Na_2SO_4 (Anhydrous) magnesium sulfate / MgSO_4 Allow silica gel / CaSO_4 Do not award anhydrous CuSO_4 / NaHSO_4 / CaCO_3 / NaOH / KOH / SiO_2 / Na_2CO_3 / NaHCO_3 If name and formula are given both must be correct (1) Mix / shake / swirl / wait until it goes clear Allow until drying agent stops lumping together Ignore "adding to mixture" Ignore "allow to react" Ignore "leave for a period of time" (1) Decant (the liquid) Allow pour off (the liquid) Allow filter (off the solid) Do not award "dry between filter paper" or "blot" Marks are independent	(3)

Question Number	Answer	Additional Guidance	Mark						
2(d)(i)	All three correct scores 2 (2)	<table><tr><td></td><td></td><td></td></tr><tr><td>Flammable</td><td>Corrosive</td><td>Irritant/Harmful/ Moderate Hazard</td></tr></table> Two correct labels in boxes scores 1 mark Allow inflammable / highly flammable Ignore “burning”, “fire” Ignore damage to skin Allow “hazardous” for exclamation mark symbol Ignore “caution” Do not award the labels for other hazard symbols e.g. “health hazard”				Flammable	Corrosive	Irritant/Harmful/ Moderate Hazard	(2)
									
Flammable	Corrosive	Irritant/Harmful/ Moderate Hazard							

Question Number	Answer	Additional Guidance	Mark
2(d)(ii)	a suitable precaution	e.g. Wear gloves Allow use small amounts Allow use a test tube rack/holder Allow keep lids on corrosive liquids (when not in use) Allow positive actions to prevent drips getting on bench e.g. place used pipettes in beaker, keep geraniol container in large beaker Ignore using a fume cupboard / wearing a face mask / lab coat / safety spectacles / clamps Ignore labelling of container Do not award dilution / decrease concentration	(1)

Question Number	Answer	Additional Guidance	Mark
2(e)	a suitable observation	Smoky / sooty flame Allow yellow / orange flame Allow black smoke Ignore black solid Do not award any other colours Ignore comments on size or luminosity of flame	(1)

Question Number	Answer	Additional Guidance	Mark
2(f)(i)	nickel / Ni (catalyst at 170°C)	Platinum / Pt OR palladium / Pd (at room temperature) Ignore temperature/heat/reflux Ignore pressure Do not award distil	(1)

Question Number	Answer	Additional Guidance	Mark
2(f)(ii)	skeletal formula	 Ignore bond lengths and angles Ignore other products and labels	(1)

(Total for Question 2 = 16 marks)

Question Number	Answer	Additional Guidance	Mark
3(a)	- axes correct way round and linear scales allow data to occupy more than half of each axis (1) - axes labelled with units (1) - all points plotted correctly (1)	An example of a graph: Temperature / °C Time / min Allow T for temperature and t for time for M2, but not T and T NB Lines not needed for (a)	(3)

Question Number	Answer	Additional Guidance	Mark
3(b)	An answer that makes reference to the following points: - two correct extrapolated lines drawn (1) - correct value for ΔT (1)	(see graph above for lines) One line is horizontal 0 to 2.5 mins, the other line is diagonal through the final points and extrapolated back to 2.5 mins. Vertical line is not required. Ignore longer extrapolated lines $\Delta T = 38.6 - 21.0$ $= 17.6\text{ }(^{\circ}\text{C})$ M2 dependent on the temperature difference being measured at 2.5 mins (Allow answers in the range 17.1 – 18.2) Allow TE from graph for M2	(2)

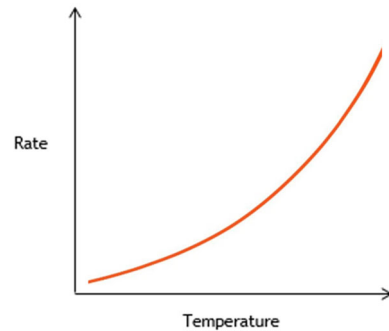
Question Number	Answer	Additional Guidance	Mark
3(c)	reason for lower value	Heat loss (to the surroundings) Heat loss (to the apparatus) Mass of solution is more than 25 g Density is more than 1 g cm^{-3} Specific heat capacity is not $4.2 / 4.18\text{ J g}^{-1}\text{ }^{\circ}\text{C}^{-1}$ Heat capacity of the polystyrene cup assumed to be 0 Allow energy loss in place of heat loss Ignore heat loss to the thermometer Ignore non-standard conditions Do not award incomplete reaction Do not award transfer errors	(1)

Question Number	Answer	Additional Guidance	Mark
3(d)(i)	• $-44.6 \text{ (kJ mol}^{-1}\text{)}$	Example of a calculation: $-39.0 - (+5.6) = -44.6 \text{ (kJ mol}^{-1}\text{)}$ Ignore units even if incorrect	(1)

Question Number	Answer	Additional Guidance	Mark
3(d)(ii)	• suitable suggestion	e.g. It is hard to add the correct amount of water e.g. Some crystals would be dissolved whilst others may not be (fully) hydrated e.g. It is hard to measure the temperature (change) of a solid Ignore copper sulfate is soluble in water "Because it is a solid" is not enough Ignore standard conditions Ignore "it is not possible to measure it"	(1)

(Total for Question 3 = 8 marks)

Question Number	Answer	Additional Guidance	Mark
4(a)(i)	An answer that makes reference to the following points: • calculation of rate (1) • answer to 1 or 2 SF (1) • units (1)	Example of a calculation: Time read from graph = 33 seconds $1 \div 33 = 0.0303$ Allow answer left as fraction 0.030 / 0.03 TE from M1 for values between 32.5 and 33.5 s^{-1} Allow "per second" Allow sec^{-1} / $seconds^{-1}$ Allow "/s" Marks are independent	(3)

Question Number	Answer	Additional Guidance	Mark
4(a)(ii)	• line that shows rate increasing with temperature AND line is curved with the gradient increasing	Example of a graph:  Allow a graph starting from the origin	(1)

Question Number	Answer	Additional Guidance	Mark
4(b)	An explanation that makes reference to the following points: - all but one point is on the best fit line / there is one anomaly (at 40°C) / a clear trend can be seen - it is not necessary to repeat the experiment as the anomaly has been identified (and excluded from the line of best fit)	(1) Allow all but one point follow a pattern Allow reference to point at 40° not being correct Allow outlier in place of anomaly (1) Allow not necessary to repeat the experiment as the pattern between rate and temperature is clear Allow not necessary to repeat as results are consistent Ignore comments about accuracy	(2)

Question Number	Answer	Additional Guidance	Mark
4(c)	An answer that makes reference to the following points: reduce the concentration (of one or more of the reactants)	Allow specific suggestions e.g. doubling/increasing volume, use a thinner/paler cross, dilute the solution Allow amount for volume Ignore pressure Ignore suggestions related to maintaining temperature at 22°C Do not award "reduce the concentration of the opaque solution" Do not award "use a different type of opaque solution"	(1)

(Total for Question 4 = 7 marks)

Question Number	Answer	Additional Guidance	Mark																														
5(a)	table completed correctly	Example table: <table><tr><td></td><td></td><td colspan="4">Titration number</td></tr><tr><td></td><td>Rough</td><td>1</td><td>2</td><td>3</td><td>4</td></tr><tr><td>Final reading / cm³</td><td>24.90</td><td>21.25</td><td>42.85</td><td>21.80</td><td>43.15</td></tr><tr><td>Initial reading / cm³</td><td>2.30</td><td>0.00</td><td>21.25</td><td>0.50</td><td>21.80</td></tr><tr><td>Titre / cm³</td><td>22.6(0)</td><td>21.25</td><td>21.6(0)</td><td>21.3(0)</td><td>21.35</td></tr></table>			Titration number					Rough	1	2	3	4	Final reading / cm ³	24.90	21.25	42.85	21.80	43.15	Initial reading / cm ³	2.30	0.00	21.25	0.50	21.80	Titre / cm ³	22.6(0)	21.25	21.6(0)	21.3(0)	21.35	(1)
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Titre / cm ³	22.6(0)	21.25	21.6(0)	21.3(0)	21.35																												

Question Number	Answer	Additional Guidance	Mark
5(b)(i)	not concordant OR more than ($\pm$)0.20 / 0.10 (cm³) from results 1, 3 and 4	Accept "Only 1, 3, and 4 are concordant / within 0.2 / 0.1 (cm ³)"	(1)

Question Number	Answer	Additional Guidance	Mark
5(b)(ii)	- calculation of mean (1) - calculation of moles of hydrochloric acid (1) - calculation of moles of sodium hydroxide solution (1) - calculation of concentration of sodium hydroxide solution (1)	An example of a calculation: $\frac{21.25 + 21.30 + 21.35}{3} = 21.3(0) \text{ (cm}^3\text{)}$ $n = c \times v = (21.30 \div 1000) \times 0.5 = 0.01065 / 1.065 \times 10^{-2}$ $0.01065 / 1.065 \times 10^{-2}$ (1:1 stoichiometry) $c = 0.01065 \div 0.025 = 0.426 / 0.43 \text{ (mol dm}^{-3}\text{)}$ Ignore SF except 1SF TE throughout Correct answer with no working scores (4) 0.587 (mol dm⁻³)/ 0.59 (mol dm⁻³) scores (3) Ignore units even if incorrect	(4)

Question Number	Answer	Additional Guidance	Mark
5(c)	(±)0.468% / (±)0.47% / (±)0.5%	An example of a calculation: $((0.05 \times 2) \div 21.35) \times 100 = (\pm)0.468\%$ Allow 1, 2 or 3SF Ignore missing percentage sign Do not award 4SF Penalise incorrect rounding	(1)

Question Number	Answer	Additional Guidance	Mark
5(d)	An answer that makes reference to the following points: • correct start colour (1) • correct end colour (1)	(Pale) pink Do not award purple Colourless Allow 1 mark for colours in reverse order	(2)

(Total for Question 5 = 9 marks)

TOTAL FOR PAPER = 50 MARKS